

**FINAL PROJECT
MARKETING DATA ANALYTICS**

SnapCash as a Transaction Solution for MSMEs



Prepared by Group 3:

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**Information Systems Project Management
Bina Nusantara University
Even Semester 2025/2026**



Aplikasi Kasir untuk para Entrepreneur



Yuk Coba Fitur kami! Bisa Scan Disini →





Fitur Inventory Management di Aplikasi Kasir untuk para Entrepreneur



Yuk Coba Fitur kami! Bisa Scan Disini →





SNAPCASH

Create a better future for SMEs.



WA : 0814-1224-8858 (Marvel)

WA : 0811-2387-300 (Tracy)

WA : 0857-5452-7330 (Cecy)

Pricing Options

⊕ Standard Package

Perfect for growing businesses seeking for efficient cashier management

Rp 100.000/month

- Cashier systems
- Inventory management systems
- Employee & owner features

⊕ Premium Package

Designed for established businesses aiming for comprehensive POS management.

Rp 300.000/month

- Cashier systems
- Inventory management systems
- Employee & owner features
- Supply Chain management
- Customized POS management Strategy

⊕ Enterprise Package

Tailored for large businesses requiring extensive business management and dedicated support.

Rp 1.500.000/month

- Cashier systems
- Inventory management systems
- Employee & owner features
- Supply Chain management
- Customized POS management Strategy
- Branch per branch integrated management POS system

*price are outside of tax and hardware price

I. Background

Indonesia's Micro, Small, and Medium Enterprises (MSMEs) serve as one of the most critical pillars of the national economy, yet many continue to operate with largely manual systems that limit their capacity for growth and data-driven decision-making. This is particularly evident in the Food and Beverage sector, where daily operations such as sales recording, stock monitoring, and financial reporting are still managed through handwritten logs or disconnected spreadsheets. The absence of an integrated digital system creates recurring inefficiencies, increases the risk of human error, and makes it difficult for business owners to gain a clear picture of their financial performance at any given time.

The idea for SnapCash emerged directly from this real-world gap. One of the founding team members observed firsthand how a family-owned FnB business struggled to manage transactions and inventory without the support of any digital tool. This experience, combined with the team's academic background in business and technology at Binus University, led to the creation of a digital Point of Sale solution designed specifically for the Indonesian MSME market. Rather than adapting an existing enterprise system, SnapCash was built from the ground up with the needs of small business owners in mind, prioritizing simplicity, affordability, and practical utility.

II. Company Description

In the rapidly growing era of digitalization in Indonesia, Micro, Small, and Medium Enterprises (MSMEs) play a vital role as a pillar of the national economy and require technological innovation to improve their operational efficiency and competitiveness. One of the main challenges faced by MSMEs, particularly in the Food and Beverage (FnB) sector, is that operational management and transactions are still largely carried out manually. This results in inefficiencies in sales recording, inventory management, and financial reporting, which ultimately limits data-driven strategic decision-making.

To address these needs, SnapCash emerges as a digital Point of Sale (POS) solution that is not only practical and affordable, but also equipped with advanced technology such as inventory management system, financial reporting, and transactional management. SnapCash is developed by a team of students from Binus University who possess strong academic backgrounds and practical experience in business and technology. The initial idea originated from a real-world need within the family of one of the founders, who still relied on manual processes to manage their FnB business. Inspired by these challenges and driven by the desire to deliver a socially impactful digital solution, SnapCash was created as a digital cashier platform that integrates transaction management, inventory control, and business analytics into a single intuitive application.

SnapCash's primary mission is to empower Indonesian MSMEs through accessible, efficient, and impactful digital technology. The tagline "Quick, Easy, Snap!" encapsulates SnapCash's three core pillars: speed in transaction processing, ease of use for users without a technical background, and modern capabilities in delivering instant and interactive data.

II. Product Backlog

ID	Backlog Item	Description	Priority	Target Period
PB-01	UI/UX Design Prototype	Design POS-friendly and cashier-optimized interface	High	March (Week 1)
PB-02	Database Design & Setup	Database for users, products, transactions, inventory, finance	High	March (Week 1)
PB-03	User Authentication & Roles	Login using email/username with role-based access (Admin, Cashier, Owner)	High	March (Week 1)
PB-04	POS Transaction Module	Process sales transactions, calculate totals, taxes, and discounts	High	March (Week 2-3)
PB-05	Receipt Generation	Generate and print/digital receipts after transactions	High	March (Week 3)
PB-06	Product Management	Add, edit, delete products with price and SKU	High	March (Week 4)
PB-07	Inventory Management	Track stock in/out automatically based on transactions	High	April (Week 1)
PB-08	Low Stock Alert	Notify admin when inventory reaches minimum threshold	Medium	April (Week 1–2)
PB-09	Supplier Management	Manage supplier data for inventory restocking	Medium	April (Week 2)
PB-10	Purchase & Restock Module	Record inventory restock and update stock levels	High	April (Week 2–3)
PB-11	Financial Report – Sales	Generate daily, weekly, and monthly sales reports	High	April (Week 3)
PB-12	Financial Report – Profit & Loss	Automatically calculate revenue, cost, and profit	High	April (Week 4)

PB-13	Cash Flow Report	Track cash inflow and outflow	Medium	May (Week 1)
PB-14	Dashboard Analytics	Visual dashboard showing sales, stock, and profit summary	High	May (Week 1–2)
PB-15	Export Reports	Export reports to PDF/Excel	Medium	May (Week 2)
PB-16	Security & Data Validation	Secure transactions, input validation, access control	Medium	May (Week 3)
PB-17	System Testing & Bug Fixing	Integration testing and bug fixing	High	May (Week 3–4)
PB-18	Deployment & Documentation	Deploy system and prepare user documentation	Medium	May (Week 4)

III. Business Case

SnapCash is a digital POS and business management platform that integrates transaction processing, inventory control, financial reporting, and business analytics into a single, accessible application. Its tagline, "Quick, Easy, Snap," reflects the three core principles that guide its design: speed in processing transactions, ease of use for operators without a technical background, and modern capability in delivering real-time, interactive business data. The system supports role-based access for owners, cashiers, and administrators, allowing businesses to manage both front-end and back-end operations from one platform.

The product is offered across three subscription tiers. The Standard Package, priced at Rp 100,000 per month, covers cashier functions, inventory management, and employee and owner features. The Premium Package at Rp 300,000 per month adds supply chain management and a customized POS strategy. The Enterprise Package at Rp 1,500,000 per month extends this further with branch-level integrated management, making it suitable for businesses with multiple outlets. This tiered structure allows SnapCash to serve a wide range of MSME segments, from a single-outlet warung to a growing multi-branch restaurant chain.

From a financial standpoint, the business case for SnapCash is grounded in the economics of a SaaS model applied to a large and underserved market. With an average revenue per user of Rp 150,000 per month, the company projects reaching 100 active customers in its first year, generating approximately Rp 180,000,000 in revenue. While the first year is expected to yield a net loss of Rp 90,000,000 due to development, infrastructure, and marketing costs, break-even is projected in the second year as the customer base scales to 300 users across expanded cities including Bandung, Surabaya, and Yogyakarta. By the third year, with 1,000

active customers and the introduction of AI-based features such as sales forecasting and inventory prediction, the platform is projected to generate a net profit of Rp 950,000,000. This trajectory continues through year five, where 3,000 active customers and a diversified revenue base that includes premium analytics, payment integrations, and business consulting services are expected to yield annual revenue exceeding Rp 5,400,000,000 and a net profit above Rp 3,200,000,000.

The long-term vision for SnapCash extends beyond being a transactional tool. By the fifth year of operation, the platform aims to position itself as a business intelligence partner for Indonesian MSMEs, exploring integration with banks, fintech providers, and MSME lending institutions. The five-year roadmap also includes the development of an open API ecosystem and marketplace integration, laying the groundwork for potential expansion into the broader ASEAN market. These projections are deliberately conservative and aligned with realistic market conditions in Indonesia, making the case for SnapCash not only as a student project, but as a commercially viable digital solution with meaningful social impact.

IV. Project Charter

A. Project General Information

Project Name

SnapCash – POS and Business Management System

Project Description

SnapCash is a digital Point of Sale (POS) system designed to help small and medium-sized businesses, especially in the food and retail sectors, manage their daily operations efficiently. The system provides features such as transaction processing, inventory management, financial reporting, and business analytics in one integrated platform. By using SnapCash, businesses can digitize their operations, reduce manual errors, and gain insights into their financial performance.

Project Duration

1 March 2026 – 31 May 2026

Total Sprint

6 Sprints (Each Sprint = 2 Weeks)

B. Role Assigning

Product Owner: Angelo

Scrum Master: Kennes Jansen

Development Team

- Christofer Marveleous Krisdianto

- Nabila Ullivia Fatimah
- Meini Rusiadi

C. Sprint Plan

Sprint 1

Duration:

1 March – 14 March 2026

Sprint Goal:

Establish system foundation including UI design, database setup, and authentication.

Sprint Backlog

ID	Task	Story Points	Assigned To
PB-01	UI/UX Design Prototype	5	Nabila Ullivia Fatimah, Meini Rusiadi
PB-02	Database Design & Setup	5	Christofer Marveleous Krisdianto
PB-03	User Authentication & Roles	5	Christofer Marveleous Krisdianto

Sprint 2

Duration:

15 March – 28 March 2026

Sprint Goal:

Develop the core POS transaction system.

Sprint Backlog

ID	Task	Story Points	Assigned to
PB-04	POS Transaction Module	8	Christofer Marveleous Krisdianto
PB-05	Receipt Generation	3	Meini Rusiadi
PB-06	Product Management	5	Nabila Ullivia Fatimah

Sprint 3

Duration:

29 March – 11 April 2026

Sprint Goal:

Develop inventory tracking and supplier management.

Sprint Backlog

ID	Task	Story Points	Assigned to
PB-07	Inventory Management	5	Christofer Marveleous Krisdianto
PB-08	Low Stock Alert	3	Nabila Ullivia Fatimah
PB-09	Supplier Management	8	Christofer Marveleous Krisdianto, Nabila Ullivia Fatimah

PB-1 0	Purchase & Restock Module	5	Meini Rusiadi
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Sprint 4

Duration:

12 April – 25 April 2026

Sprint Goal:

Develop financial reporting features.

Sprint Backlog

ID	Task	Story Points	Assigned to
PB-11	Financial Report – Sales	8	Christofer Marveleous Krisdianto
PB-12	Financial Report – Profit & Loss	5	Meini Rusiadi
PB-13	Cash Flow Report	5	Nabila Ullivia Fatimah

Sprint 5

Duration:

26 April – 9 May 2026

Sprint Goal:

Develop business analytics and report exporting features.

Sprint Backlog

ID	Task	Story Points	Assigned to
PB-14	Dashboard Analytics	5	Meini Rusiadi
PB-15	Export Reports	5	Nabila Ullivia Fatimah
PB-16	Security & Data Validation	5	Christofer Marveleous Krisdianto

Sprint 6

Duration:

10 May – 23 May 2026

Sprint Goal:

Finalize system development and prepare deployment.

Sprint Backlog

ID	Task	Story Points	Assigned to
PB-1 7	System Testing & Bug Fixing	8	Christofer Marveleous Krisdianto, Meini Rusiadi
PB-1 8	Deployment & Documentation	5	Nabila Ullivia Fatimah

E. Final Phase

24 May – 31 May 2026

Activities:

- Final system testing

- Deployment preparation
- Use documentation creation
- Project evaluation
- Final presentation preparation

Definition of Done (DoD)

A backlog item is considered complete when:

- The feature has been implemented successfully
- The feature passes testing without major bugs
- The system integrates with other modules correctly
- Code has been committed to the repository
- The feature is ready to be demonstrated in Sprint Review

Sprint Execution Plan – SnapCash

Sprint 1

Duration: 1 March – 14 March 2026

Sprint Focus

Establish the **foundation of the system**, including UI/UX design, database setup, and user authentication.

Activities

- Design POS cashier-friendly interface prototype
- Create database schema for users, products, transactions, inventory, and finance
- Setup development environment and repository
- Implement login system with role-based access (Admin, Cashier, Owner)
- Integrate authentication with database

Deliverables

- UI/UX POS Prototype
- Database structure and tables
- Functional login system with role-based access
- Initial system architecture ready for development

Sprint 2

Duration: 15 March – 28 March 2026

Sprint Focus

Develop the **core POS transaction system** and product management features.

Activities

- Implement POS transaction processing module
- Develop product management feature (add/edit/delete products)
- Integrate product database with POS system
- Implement receipt generation for completed transactions
- Conduct initial testing for transaction flow

Deliverables

- Functional POS transaction module
- Product management system
- Digital receipt generation
- Tested transaction workflow

Sprint 3

Duration: 29 March – 11 April 2026

Sprint Focus

Develop **inventory management and supplier management features**.

Activities

- Implement inventory tracking system
- Integrate automatic stock deduction during transactions
- Develop low stock alert notification system
- Create supplier management module
- Develop purchase and restock module for inventory updates

Deliverables

- Inventory management system
- Low stock alert feature
- Supplier management module
- Purchase and restock functionality

Sprint 4

Duration: 12 April – 25 April 2026

Sprint Focus

Develop **financial reporting features** to track business performance.

Activities

- Implement daily, weekly, and monthly sales reports
- Develop profit and loss calculation system
- Implement cash flow tracking functionality
- Integrate financial data with transaction and inventory systems
- Test report accuracy and data consistency

Deliverables

- Sales report system
- Profit and Loss reporting feature
- Cash flow report feature
- Integrated financial reporting module

Sprint 5

Duration: 26 April – 9 May 2026

Sprint Focus

Develop **business analytics dashboard and reporting features**.

Activities

- Design and develop dashboard analytics interface
- Implement sales, inventory, and financial summary visualization
- Develop export functionality for reports (PDF/Excel)
- Implement security features and input validation
- Improve system usability and user interface

Deliverables

- Dashboard analytics interface
- Exportable reports (PDF/Excel)
- Security and data validation system
- Improved user interface and usability

Sprint 6

Duration: 10 May – 23 May 2026

Sprint Focus

Finalize system development through **testing, debugging, deployment preparation, and documentation.**

Activities

- Conduct full system integration testing
- Identify and fix system bugs and errors
- Optimize system performance
- Prepare deployment environment
- Create user documentation and system guide

Deliverables

- Fully integrated SnapCash system
- Bug fixes and optimized system performance
- Deployed system ready for use
- User documentation and deployment report

Final Project Activities

24 May – 31 May 2024

Activities:

- Final system review
- Product demonstration
- Documentation completion
- Final presentation preparation

Deliverables:

- Final SnapCash system
- Complete project documentation
- Final project presentation

Daily Scrum Log – SnapCash Project

Daily Scrum Duration: 45 Minutes

Meeting Time: 09.00 AM – 09.45 AM

Participants:

- Product Owner
- Scrum Master
- Development Team

Sprint 1 Daily Scrum Log

Sprint Duration: 1 – 14 March 2026

Sprint Focus: System foundation (UI/UX, Database, Authentication)

Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning and requirement discussion	Setup project repository and development tools	None
Day 2	Created initial UI wireframe	Continue POS UI prototype design	None
Day 3	Completed POS interface design draft	Begin database schema design	None
Day 4	Designed database schema	Implement database tables	Minor schema adjustments
Day 5	Database setup completed	Start login authentication system	None
Day 6	Developed login interface	Implement role-based access control	None

Day 7	Integrated login system with database	Testing authentication system	None
Day 8	Fixed login validation issues	Prepare system for next module	None
Day 9	Final authentication testing	Documentation for sprint deliverables	None
Day 10	Sprint review preparation	Sprint retrospective	None

Sprint 2 Daily Scrum Log

Sprint Duration: 15 – 28 March 2026

Sprint Focus: POS transaction and product management

Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design POS transaction flow	None
Day 2	POS workflow designed	Implement transaction logic	None
Day 3	Transaction module development	Continue transaction processing	None
Day 4	Product database integration	Build product management interface	None

Day 5	Product CRUD feature implemented	Integrate POS with product list	None
Day 6	Transaction testing	Implement receipt generation	None
Day 7	Receipt module development	Continue receipt formatting	None
Day 8	Transaction & receipt integration	Fix calculation issues	Minor bugs
Day 9	System testing	Prepare sprint deliverables	None
Day 10	Sprint review preparation	Sprint retrospective	None

Sprint 3 Daily Scrum Log

Sprint Duration: 29 March – 11 April 2026

Sprint Focus: Inventory and supplier management

Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design inventory structure	None
Day 2	Inventory database tables	Implement stock tracking logic	None
Day 3	Automatic stock deduction feature	Testing inventory updates	None

Day 4	Inventory testing	Implement low stock alert	None
Day 5	Low stock alert notification	Supplier management design	None
Day 6	Supplier database setup	Supplier CRUD features	None
Day 7	Purchase/restock module	Integrate supplier system	None
Day 8	Inventory restock testing	Fix stock update bugs	Minor bug
Day 9	System integration testing	Documentation	None
Day 10	Sprint review preparation	Sprint retrospective	None

Sprint 4 Daily Scrum Log

Sprint Duration: 12 – 25 April 2026

Sprint Focus: Financial reporting system

Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design reporting structure	None
Day 2	Sales report development	Integrate transaction data	None

Day 3	Sales report completed	Develop profit & loss module	None
Day 4	Profit & loss calculations	Testing financial accuracy	None
Day 5	Fix calculation errors	Develop cash flow report	None
Day 6	Cash flow module development	Integrate reports with dashboard	None
Day 7	Report interface improvements	Testing report generation	None
Day 8	Financial report debugging	Data validation	None
Day 9	System testing	Prepare sprint review	None
Day 10	Sprint review preparation	Sprint retrospective	None

Sprint 5 Daily Scrum Log

Sprint Duration: 26 April – 9 May 2026

Sprint Focus: Dashboard analytics and reporting export

Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design dashboard UI	None

Day 2	Dashboard prototype	Implement data visualization	None
Day 3	Sales analytics chart	Integrate inventory analytics	None
Day 4	Dashboard integration	Develop export report feature	None
Day 5	PDF export system	Excel export system	None
Day 6	Export testing	Implement security validation	None
Day 7	Access control improvements	Input validation testing	None
Day 8	Dashboard usability improvements	Fix UI bugs	Minor UI issues
Day 9	Final dashboard testing	Documentation	None
Day 10	Sprint review preparation	Sprint retrospective	None

Sprint 6 Daily Scrum Log

Sprint Duration: 10 – 23 May 2026

Sprint Focus: System testing, deployment, documentation

Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Begin system integration testing	None
Day 2	Integration testing	Identify system bugs	None
Day 3	Fix transaction module bugs	Continue debugging	None
Day 4	Fix inventory bugs	Optimize system performance	None
Day 5	Performance testing	Security testing	None
Day 6	Security improvement	Prepare deployment environment	None
Day 7	Deployment preparation	Create user documentation	None
Day 8	Documentation writing	Final system testing	None
Day 9	Final bug fixes	Prepare final presentation	None
Day 10	Sprint review preparation	Final retrospective	None

V. SnapCash Work Breakdown Structure (WBS)

Work Breakdown Structure						
Project Title	SnapCash					
Team Leader	Angelo					
Company Name	SnapCash					
Date	28/02/2026					
WBS Number	Task Title	Persons	Start Date	Due Date	Progress	
1	PROJECT INITIATION & PLANNING					
1.1	Team Forming	All Member ▼	07/02/2026	07/02/2026	Done ▼	
1.2	Define Project Scope	All Member ▼	07/02/2026	07/02/2026	Done ▼	
1.3	Define Project Objectives	Angelo ▼	07/02/2026	07/02/2026	Done ▼	
Deliverables	Project Scope and Objectives					
2	REQUIREMENTS GATHERING & ANALYSIS					
2.1	System Architecture Analysis	Abil ▼	07/02/2026	20/02/2026	Done ▼	
2.2	UI/UX Prototyping	Meini ▼	01/03/2026	07/03/2026	Done ▼	
2.3	Database Design and Setup	Kennes ▼	01/03/2026	08/03/2026	Done ▼	
2.4	Define Functional Requirements	Kennes ▼	01/03/2026	08/03/2026	Done ▼	
Deliverables	System Architecture, UI/UX Prototype, Database Setup, Functional Requirements					
3	SYSTEM & DATA MODELLING					
3.1	User Authentication and Roles	Marvel ▼	01/03/2026	08/03/2026	To Do ▼	
3.2	POS Transaction Module	Marvel ▼	09/03/2026	23/03/2026	To Do ▼	
3.3	Receipt Generation	Marvel ▼	16/03/2026	23/03/2026	To Do ▼	
3.4	Product Management	Marvel ▼	24/03/2026	31/03/2026	To Do ▼	
3.5	Inventory Management	Kennes ▼	01/04/2026	07/04/2026	To Do ▼	
3.6	Low Stock Alert	Kennes ▼	08/04/2026	22/04/2026	To Do ▼	
3.7	Supplier Management	Kennes ▼	08/04/2026	15/04/2026	To Do ▼	
3.8	Purchase & Restock Module	Marvel ▼	08/04/2026	22/04/2026	To Do ▼	
3.9	Financial Report - Profit and Loss	Marvel ▼	16/04/2026	22/04/2026	To Do ▼	
3.10	Financial Report- Sales	Abil ▼	22/04/2026	30/04/2026	To Do ▼	
3.11	Cashflow Report	Abil ▼	01/05/2026	08/05/2026	To Do ▼	
3.12	Dashboard Analytics	Meini ▼	01/05/2026	15/05/2026	To Do ▼	
3.13	Export Reports	Meini ▼	08/05/2026	15/05/2026	To Do ▼	
Deliverables	Usable and Ready to Test SnapCash App				To Do ▼	
6	Testing					
4.1	Security & Data validation	Angelo ▼	16/05/2026	23/05/2026	To Do ▼	
4.2	Final Integration Testing & Bug Fixing	Angelo ▼	16/05/2026	31/05/2026	To Do ▼	
4.3	Deployment & Documentation	Angelo ▼	24/05/2026	31/05/2026	To Do ▼	
Deliverables	Ready to use SnapCash App					

The Work Breakdown Structure is a hierarchical decomposition of the total scope of work required to complete the SnapCash project. It divides the project into four major phases, each representing a distinct stage of development. Every phase contains specific tasks assigned to individual team members, along with defined start dates, due dates, and expected deliverables. The WBS serves as the foundation for planning, scheduling, and monitoring the project.

Phase 1: Project Initiation & Planning (WBS 1)

This is the first and most fundamental phase of the project. It covers the initial activities required to formally start the project, including forming the team, defining the project scope, and establishing the project objectives. All three tasks were assigned to all team members and Angelo, and were completed on February 7, 2026 — the very first day of the project. The deliverable for this phase is the Project Scope and Objectives document, which serves as the guiding reference for all subsequent phases. The rapid completion of this phase on a single day reflects its nature as a planning sprint rather than a prolonged activity.

Phase 2: Requirements Gathering & Analysis (WBS 2)

The second phase focuses on understanding what the system needs to do and how it should be built. It consists of four tasks: System Architecture Analysis (assigned to Abil), UI/UX Prototyping (assigned to Meini), Database Design and Setup (assigned to Kennes), and Define Functional Requirements (also assigned to Kennes). This phase began on February 7, 2026, and concluded on March 8, 2026. The deliverables include the System Architecture document, UI/UX Prototype, Database Setup, and Functional Requirements specification. All four tasks have been marked as Done, meaning the team has successfully completed the analysis and design groundwork needed before development begins.

Phase 3: System & Data Modelling (WBS 3)

This is the largest and most complex phase of the project, comprising thirteen individual tasks that cover the core development of the SnapCash application. The tasks span from March 1, 2026 to May 15, 2026 and are distributed among four team members: Marvel, Kennes, Abil, and Meini. Marvel is responsible for building the foundational system modules: User Authentication and Roles, POS Transaction Module, Receipt Generation, Product Management, and the financial reporting modules including Financial Report - Profit and Loss. Kennes handles the inventory and operations side, covering Inventory Management, Low Stock Alert, and Supplier Management, as well as the Purchase and Restock Module. Abil is responsible for the Financial Report - Sales and Cashflow Report modules. Meini is tasked with the Dashboard Analytics and Export Reports features. All thirteen tasks in this phase are currently marked as To Do, indicating that the development work is yet to begin. The deliverable upon completion of this phase is a usable and ready-to-test SnapCash application.

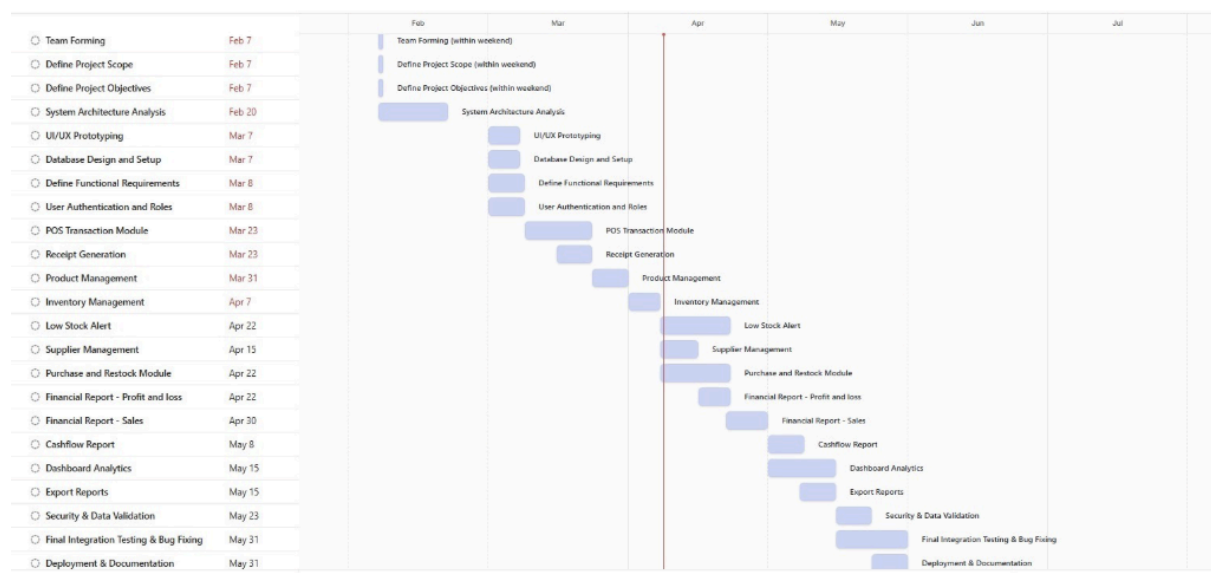
Phase 4: Testing & Deployment (WBS 6)

The fourth and final phase covers the quality assurance and release activities for the project. It contains three tasks, all assigned to Angelo: Security and Data Validation, Final Integration Testing and Bug Fixing, and Deployment and Documentation. This phase is scheduled to run from May 16 to May 31, 2026. All three tasks are currently marked as To Do. The deliverable of this phase is the final, ready-to-use SnapCash application. This phase is critical because it ensures the system is secure, fully functional, and properly documented before it is handed over for actual use.

WBS Summary Table

WBS No.	Phase / Task	Person(s)	Start Date	Due Date	Status
1	Project Initiation & Planning	All Members	07/02/2026	07/02/2026	Done
2	Requirements Gathering & Analysis	Abil, Meini, Kennes	07/02/2026	08/03/2026	Done
3	System & Data Modelling	Marvel, Kennes, Abil, Meini	01/03/2026	15/05/2026	To Do (13 tasks)
4 (6)	Testing & Deployment	Angelo	16/05/2026	31/05/2026	To Do (3 tasks)

SnapCash Gantt Chart



The Gantt Chart is a visual timeline that maps each task against a calendar spanning from February 2026 to May 2026. It provides a clear view of the duration, sequencing, and overlap of tasks across the project. The chart lists all twenty tasks on the vertical axis, while the horizontal axis represents time in months. Each task is represented by a horizontal bar whose length corresponds to the duration of that task.

Timeline Structure and Task Sequencing

Looking at the Gantt Chart, it is clear that the project follows a structured, sequential flow with some parallel activities. The earliest tasks which are Team Forming, Define Project

Scope, and Define Project Objectives are all labeled as occurring within a single weekend in early February, reflecting the rapid completion shown in the WBS. These are immediately followed by the analysis phase tasks, beginning with System Architecture Analysis, which extends from early February to February 20.

The UI/UX Prototyping, Database Design and Setup, Define Functional Requirements, and User Authentication and Roles tasks all converge around the March 7-8 timeframe, indicating a period of parallel development during the transition from requirements to system building. This overlap is intentional and reflects an efficient use of team resources, as different members work on different components simultaneously.

Development Phase Timeline

The bulk of the Gantt Chart is occupied by Phase 3 tasks, which span from early March to mid-May. The development work follows a logical progression: core system modules such as the POS Transaction Module and Receipt Generation are due by March 23, followed by Product Management due March 31. Inventory and operations modules (Inventory Management, Low Stock Alert, Supplier Management, Purchase and Restock Module) are all due in early-to-mid April. Financial reporting modules follow in late April, and the Dashboard Analytics and Export Reports tasks extend into mid-May. This sequencing ensures that foundational modules are built before more dependent features like analytics and reporting are completed.

Testing and Deployment Window

The final section of the Gantt Chart covers the testing and deployment phase from mid-May to May 31. Security and Data Validation is due May 23, while Final Integration Testing and Bug Fixing and Deployment and Documentation are both due May 31. The two-week window allocated for testing is relatively tight given the breadth of the system, which suggests that thorough testing during Phase 3 development will be important to minimize the number of issues surfaced during the dedicated testing phase.

Team Responsibilities Summary

Across both the WBS and Gantt Chart, it is evident that the workload is distributed based on each team member's area of responsibility. Marvel carries the heaviest development load, handling the core transactional and product-related modules. Kennes manages inventory and supplier operations. Abil focuses on financial reporting. Meini handles the user interface and analytics features. Angelo, as team leader, is responsible for the initiation phase and takes sole responsibility for the critical final phase of testing and deployment, ensuring accountability at both the start and end of the project.

SnapCash Project Budget Plan

POS System with Integrated Inventory Management & Financial Reporting

Total Budget: Rp 200,000,000 | Team: 5 members | Duration: 4–6 months | Type: Startup / Business

Category	Amount (IDR)	%	Breakdown
Human Resources	Rp 75,000,000	37.5%	5 developers × 5 months @ Rp3M/person/month
Infrastructure & Hosting	Rp 30,000,000	15%	Cloud server, database, domain & SSL — Rp6M/month × 5
Software & Tools	Rp 25,000,000	12.5%	Development licenses, APIs, third-party subscriptions
Hardware / Equipment	Rp 25,000,000	12.5%	POS terminal, barcode scanner, receipt printer for testing
UI/UX Design	Rp 15,000,000	7.5%	Figma licenses, design sprints, prototyping & assets
Testing & QA	Rp 10,000,000	5%	Manual & automated testing, bug fixes, UAT sessions
Documentation & Training	Rp 10,000,000	5%	User manuals, onboarding materials, handover docs
Contingency Reserve	Rp 10,000,000	5%	Buffer for unexpected costs — only use if overrun occurs
TOTAL	Rp 200,000,000	100%	Full project budget — 4 to 6 months

Budget Explanation

1. Human Resources — Rp 75,000,000 (37.5%)

The largest single allocation covers the compensation of all 5 team members across the full 5-month development period. Each member receives Rp 3,000,000 per month, reflecting a fair distribution of effort across planning, development, testing, and deployment. This cost should be tracked monthly and adjusted only if the project timeline extends beyond 5 months.

2. Infrastructure & Hosting — Rp 30,000,000 (15%)

Covers recurring cloud server costs, relational database hosting, SSL certificates, and a custom domain for SnapCash. Estimated at Rp 6,000,000 per month over 5 months. This includes development, staging, and production environments. Provider selection (e.g., AWS, Google Cloud, or a local Indonesian provider) should be finalized in Phase 1 to lock in pricing.

3. Software & Tools — Rp 25,000,000 (12.5%)

One-time and subscription-based software licenses needed during development. This includes payment gateway integration fees, barcode/QR APIs, project management tools (e.g., Jira or Notion), and code repository access. Prioritize open-source solutions where possible to reduce spend in this category.

4. Hardware / Equipment — Rp 25,000,000 (12.5%)

Allocated for the procurement of physical POS hardware required for testing and the client demo environment. This includes a POS touchscreen terminal, barcode scanner, and thermal receipt printer. Hardware should be purchased in Phase 2 once the core software features are confirmed, to avoid purchasing incompatible equipment.

5. UI/UX Design — Rp 15,000,000 (7.5%)

Funds the design sprint in Phase 1, covering Figma Pro licenses, creation of wireframes, hi-fidelity mockups, and interactive prototypes for the POS interface, inventory dashboard, and financial reporting screens. Good UI/UX investment early reduces costly redesigns during development.

6. Testing & QA — Rp 10,000,000 (5%)

Dedicated budget for systematic quality assurance in Phase 3. Covers manual testing across all modules (sales, inventory, reports), automated test scripts, and a User Acceptance Testing (UAT) session with a real or simulated end user. Any critical bugs identified during UAT should be resolved before the final handover.

7. Documentation & Training — Rp 10,000,000 (5%)

Covers the production of user-facing documentation including an admin manual, cashier quick-start guide, and system architecture notes for handover. Also includes a short training session for end users or stakeholders. This phase marks the final deliverable of the project.

8. Contingency Reserve — Rp 10,000,000 (5%)

A non-allocatable buffer reserved exclusively for cost overruns, unplanned integrations, or scope changes approved by the group. This fund must not be pre-spent or distributed early. If unused at project close, it can be returned or redirected to a post-launch support period. All use of contingency funds must be documented with justification.

Budget rule: The contingency reserve (Rp 10,000,000) must not be allocated ahead of time. All category spending should be reviewed at the end of each project phase. Any unspent funds in one category may be reallocated with full team approval.

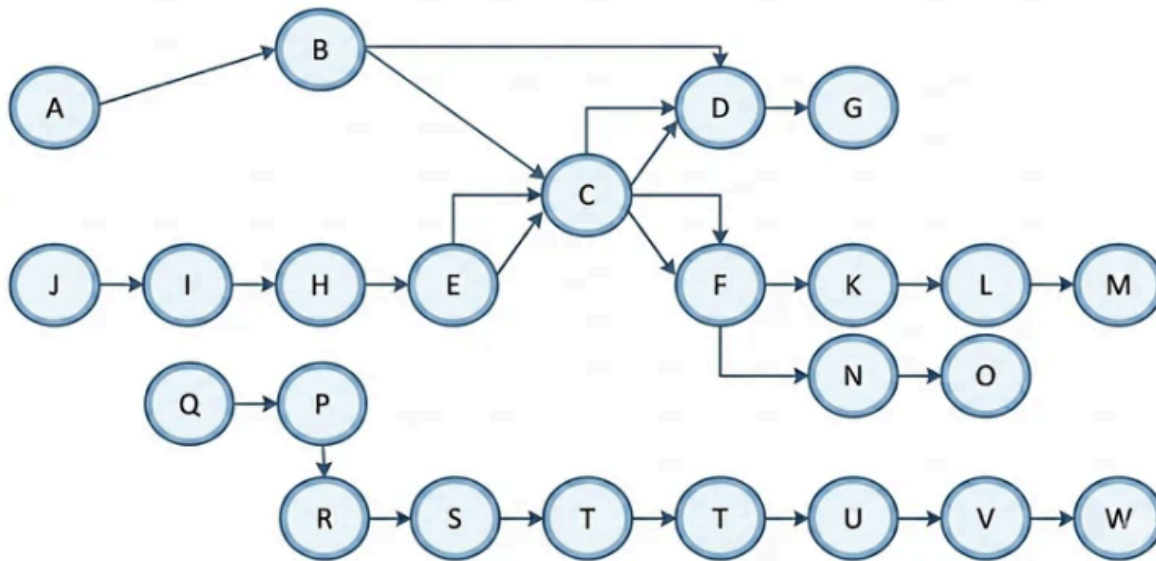
VI. Project Schedule

Activity On Node

Task	Task Title	Duration	Predecessor
A	Team Forming	1	None
B	Define Project Scope	1	A
C	Define Project Objectives	1	A
D	System Architecture Analysis	14	B,C
E	UI/UX Prototyping	7	B,C
F	Database Design and Setup	7	B,C
G	Define Functional Requirements	7	D
H	User Authentication and Roles	7	E,F
I	POS Transaction Module	14	E,F
J	Receipt Generation	7	I
K	Product Management	7	E,F
L	Inventory Management	7	K
M	Low Stock Alert	14	L
N	Supplier Management	7	E,F
O	Purchase & Restock Module	14	N
P	Financial Report - Profit and Loss	7	I
Q	Financial Report- Sales	7	I
R	Cashflow Report	7	P,Q
S	Dashboard Analytics	14	P,Q,R
T	Export Reports	7	S
U	Security & Data validation	7	T
V	Final Integration Testing & Bug Fixing	14	U
W	Deployment & Documentation	7	V

Arrow Diagram Method

Activity-on-Arrow Network Diagram



Critical Path Method

FORWARD PASS

Task	ES	EF	Ket
A	0	1	Mulai dari 0
B	1	2	EF A jadi ES B
C	1	2	EF A jadi ES C
D	2	16	Max EF B,C
E	2	9	Max EF B,C
F	2	9	Max EF B,C
G	16	23	EF D
H	9	16	Max EF E,F
I	9	23	Max EF E,F
J	23	30	EF I
K	9	16	Max EF E,F
L	16	23	EF K
M	23	37	EF L
N	9	16	Max EF E,F
O	16	30	EF N
P	23	30	EF I
Q	23	30	EF I
R	30	37	Max EF P,Q
S	37	51	Max EF R
T	51	58	EF S
U	58	65	EF T
V	65	79	EF U
W	79	86	Proyek selesai 86 HARI

BACKWARD PASS

Task	Successor	LF	LS	Ket
W	None	86	79	Start from project duration
V	W	79	65	
U	V	65	58	
T	U	58	51	
S	T	51	37	
R	S	37	30	
P	R	30	23	
Q	R	30	23	
O	None	86	72	
N	O	72	65	
M	None	86	72	
L	M	72	65	
K	L	65	58	
J	None	86	79	
I	J,P,Q	23	9	
H	None	86	79	
G	None	86	79	
F	H,I,K,N	9	2	
E	H,I,K,N	9	2	
D	G	79	65	
C	D,E,F	2	1	
B	D,E,F	2	1	
A	B,C	1	0	

VII. Risk Management Plan

1. Risk Assessment Methodology

Risks are assessed using a 3×3 Likelihood × Impact matrix. Each risk is scored and categorized into one of four levels:

Risk Level	Score	Description
CRITICAL	High × High	Immediate action required; escalate to Scrum Master
HIGH	Med-High	Develop detailed mitigation plan within current sprint
MEDIUM	Medium	Monitor weekly; address if trend worsens
LOW	Low	Accept and monitor passively

2. Risk Probability & Impact Matrix

Impact ↓ / Likelihood →	Low	Medium	High
Low	LOW R-05, R-14	LOW	MEDIUM
Medium	LOW	MEDIUM R-04, R-08 R-11, R-13	HIGH R-02, R-03 R-12, R-15
High	MEDIUM	HIGH R-10	CRITICAL R-01, R-06 R-07, R-09

3. Risk Register

ID	Category	Risk Description	Likelihood	Impact	Level	Mitigation
R-01	Technical	Integration bugs between POS, inventory, and financial modules	High	High	CRITICAL	Conduct integration testing every sprint; use a shared API contract between modules.
R-02	Technical	Database schema errors causing data loss or inconsistency	Medium	High	HIGH	Implement database migration scripts; conduct schema review before Sprint 2.
R-03	Technical	Authentication/role-based access vulnerability	Medium	High	HIGH	Use JWT token + bcrypt hashing; conduct security audit in Sprint 5 (PB-16).
R-04	Technical	Low performance when processing many concurrent transactions	Medium	Medium	MEDIUM	Optimize database queries; add indexing on critical fields; load-test before deployment.
R-05	Technical	Export report (PDF/Excel) format incompatibility	Low	Medium	LOW	Use well-tested libraries (e.g., jsPDF, SheetJS); test on multiple devices.

R-06	Schedule	Sprint delay due to underestimated story points	High	High	CRITICAL	Add buffer time each sprint; hold daily standups; re-estimate in Sprint Review if needed.
R-07	Schedule	System Testing & Bug Fixing (PB-17) underscoped, pushing beyond deadline	High	High	CRITICAL	Begin regression testing from Sprint 4 onward; not only in Sprint 6.
R-08	Schedule	Late deployment causes incomplete user documentation	Medium	Medium	MEDIUM	Prepare documentation template early; assign dedicated member (Nabila) from Sprint 5.
R-09	Team	Key developer (Christofer) overloaded – assigned to 6+ backlog items	High	High	CRITICAL	Redistribute tasks; pair-program critical features; Scrum Master monitors workload.
R-10	Team	Team member absence or dropout mid-project	Low	High	HIGH	Document all code with comments; use Git for version control; cross-train team members.

R-11	Team	Miscommunication between Product Owner and Development Team	Medium	Medium	MEDIUM	Hold Sprint Planning & Review meetings consistently; use shared backlog in project management tool.
R-12	Scope	Scope creep – adding features beyond original backlog	Medium	High	HIGH	Strictly follow Definition of Done; Product Owner (Angelo) approves all scope changes.
R-13	Scope	Dashboard Analytics (PB-14) requirements unclear or too complex	Medium	Medium	MEDIUM	Define specific KPIs/metrics for dashboard in Sprint 4; create wireframe before coding.
R-14	External	Third-party library deprecation or breaking change (e.g., POS UI library)	Low	Medium	LOW	Lock dependency versions; monitor changelog; use widely-supported libraries.
R-15	External	MSME user adoption lower than expected due to UI complexity	Medium	High	HIGH	Conduct usability testing with real MSME users in Sprint 5; iterate based on feedback.

4. Risk Ownership & Monitoring

ID	Risk	Owner	Review	Mitigation
R-01	Integration bugs between modules	Christofer (Lead Dev)	Every Sprint	Roll back to last stable version; isolate failing module.
R-02	Database schema errors	Christofer	Sprint 1–2	Restore from backup; apply migration rollback script.
R-03	Authentication vulnerability	Christofer	Sprint 5	Disable affected endpoint; hotfix within 24 hours.
R-04	Low performance on concurrent transactions	Christofer	Sprint 5–6	Reduce query complexity; cache frequently accessed data.
R-05	Export report incompatibility	Nabila	Sprint 5	Switch to alternative library; manual export fallback.
R-06	Sprint delay	Kennes (Scrum Master)	Every Sprint	Reprioritize backlog; defer low-priority items to next sprint.
R-07	Testing underscoped	Kennes (Scrum Master)	Sprint 4–6	Extend Sprint 6; reduce scope of non-critical features.
R-08	Incomplete documentation	Nabila	Sprint 5–6	Use auto-documentation tools; focus on critical user flows.
R-09	Christofer overloaded	Kennes (Scrum Master)	Every Sprint	Pair-program with Meini; redistribute tasks in Sprint Planning.

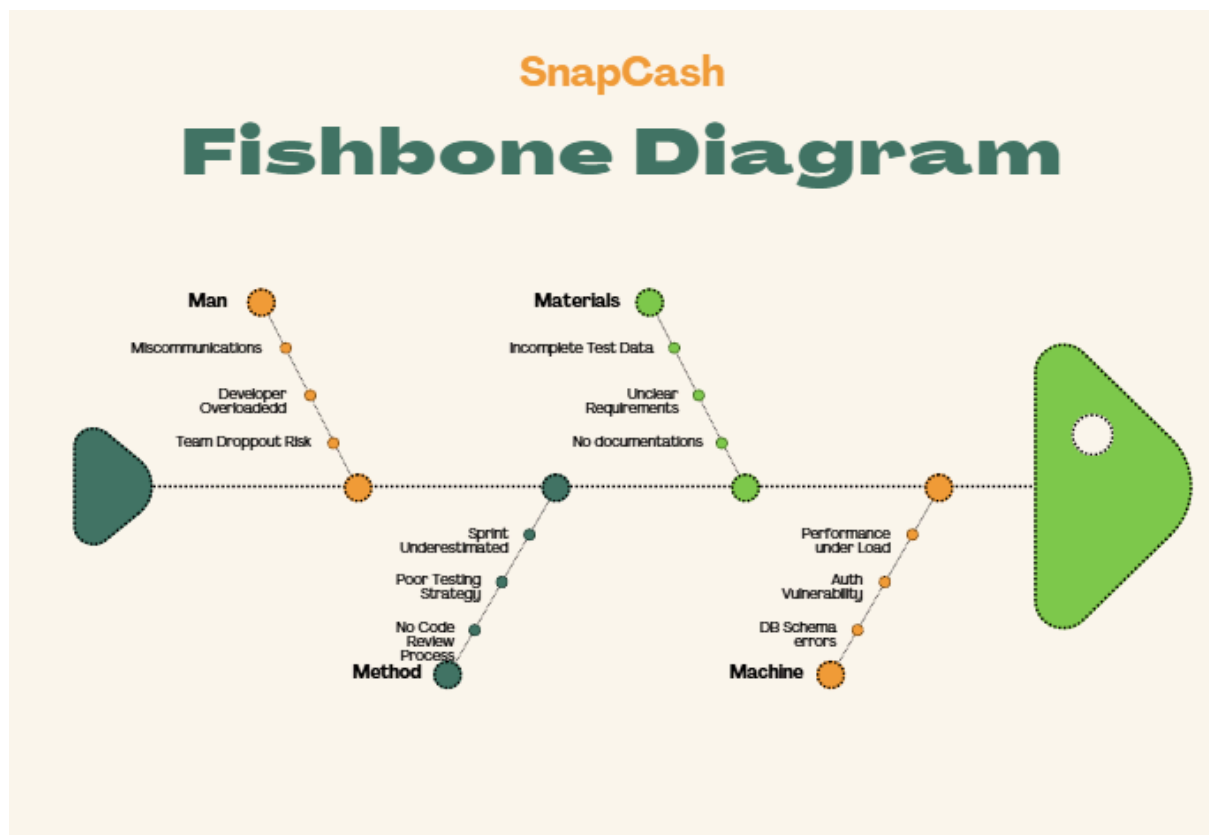
R-1 0	Team member dropout	Angelo (PO)	Ongoing	Redistribute tasks; prioritize critical features only.
R-1 1	Miscommunication PO ↔ Dev Team	Angelo (PO)	Every Sprint	Re-conduct Sprint Planning; clarify acceptance criteria.
R-1 2	Scope creep	Angelo (PO)	Every Sprint	Freeze scope; log new requests for post-project phase.
R-1 3	Dashboard Analytics unclear requirements	Meini	Sprint 4	Simplify to 3 core KPIs: daily revenue, stock level, profit.
R-1 4	Third-party library breaking change	Christofer	Sprint 2, 5	Pin dependency versions in package.json; test after updates.
R-1 5	Low MSME user adoption	Angelo (PO)	Sprint 5–6	Redesign critical workflows; provide onboarding walkthrough.

5. Risk Summary by Sprint

Sprint	Period	Key Risks to Watch	Recommended Action
Sprint 1	1–14 Mar	R-02 (DB Schema), R-03 (Auth), R-09 (Overload)	Define DB schema carefully; peer-review before finalizing.
Sprint 2	15–28 Mar	R-01 (Integration), R-06 (Delay), R-09 (Overload)	Set API contract between POS & Inventory modules early.
Sprint 3	29 Mar–11 Apr	R-01, R-06, R-12 (Scope Creep)	Strictly enforce DoD; avoid adding new inventory features.
Sprint 4	12–25 Apr	R-11 (Miscommunication), R-13 (Dashboard Scope)	Clarify financial KPIs with PO before development starts.

Sprint 5	26 Apr–9 May	R-03 (Security), R-15 (Usability), R-07 (Testing)	Start usability tests; run security audit on authentication.
Sprint 6	10–23 May	R-07 (Testing Scope), R-08 (Documentation), R-06	Prioritize bug fixes; complete documentation in parallel.

VIII. Fishbone Diagram



IX. Communication Plan

1. Purpose of the Communication Plan

The Communication Management Plan for the SnapCash project defines how information is distributed to each stakeholder throughout the project lifecycle. It ensures that every stakeholder receives the right information, in the right format, at the right time, through the appropriate contact person. The plan covers all four phases of the project: Project Initiation and Planning, Requirements Gathering and Analysis, System and Data Modelling, and Testing and Deployment.

2. Stakeholder Information Needs

Each stakeholder group has specific information requirements based on their role in the project:

Stakeholder	Information Needed	Reason
Project Sponsor / Top Management	Overall project progress, budget usage, milestone completion, risk status	To make high-level decisions and approve phase transitions
Development Team (Marvel, Kennes, Abil, Meini)	Task assignments, technical specifications, deadlines, dependency updates, bug reports	To execute development tasks on schedule and coordinate between modules
Team Leader (Angelo)	Full project status, testing results, deployment readiness, team performance	To lead and control all project phases, especially testing and deployment
End Users / Client	Feature delivery updates, UAT (User Acceptance Testing) schedule, training plan	To prepare for system adoption and provide feedback during testing
QA / Testing Team (Angelo)	Test plans, bug tracking reports, integration test results, deployment checklist	To validate system quality and sign off before go-live

3. Communication Plan: Stakeholder Analysis

Stakeholder	Document Name	Document Format	Contact Person	Due	Information Content
Phase 1: Project Initiation and Planning					
All Team Members	Project Kick-off Document	Hard copy and Meeting	Angelo (Team Leader)	February 7, 2026	Team roles, project goals, scope, timeline overview

Top Management / Sponsor	Project Scope and Objectives Report	Hard copy	Angelo (Team Leader)	February 7, 2026	Approved project scope, objectives, and initial resource plan
Phase 2: Requirements Gathering and Analysis					
Development Team	System Architecture Document	E-mail and Meeting	Nabila	February 20, 2026	System architecture design, component diagram, tech stack decisions
Development Team	UI/UX Prototype Report	E-mail	Meini	March 7, 2026	Wireframes, user flow diagrams, UI design guidelines
Development Team	Database Design Document	E-mail	Kennes	March 7, 2026	Database schema, entity relationships, data dictionary
All Team Members	Functional Requirements Specification	Hard copy and Meeting	Kennes	March 8, 2026	Full list of system features, functional and non-functional requirements
Phase 3: System and Data Modelling					

Development Team	Weekly Progress Report	E-mail	Angelo (Team Leader)	Every Monday (Mar to May)	Status of each module, blockers, percentage completion per task
Top Management	Monthly Status Report	Hard copy and Meeting	Angelo (Team Leader)	1st of each month	High-level milestone progress, risks, budget utilization, schedule adherence
Marvel (Core Dev)	Core Module Completion Report	E-mail	Christofer	March 8, 23, 31 and April 22	Auth, POS, Receipt, Product Mgmt, P&L module delivery status
Kennes (Ops Dev)	Operations Module Report	E-mail	Kennes	April 7, 15, and 22	Inventory, Stock Alert, Supplier, Restock module delivery status
Abil (Finance Dev)	Financial Report Module Update	E-mail	Nabila	April 22 and May 8	Sales and Cashflow report module progress and test results
Meini (Analytics Dev)	Analytics Module Report	E-mail	Meini	May 15	Dashboard Analytics and Export Reports completion status
Phase 4: Testing and					

Deployment					
QA / Testing Team	Security and Validation Test Report	Hard copy	Angelo	May 23, 2026	Security vulnerability results, data validation test cases and outcomes
All Team Members	Integration Test and Bug Fix Report	Hard copy and Meeting	Angelo	May 31, 2026	Bug list, severity levels, fix status, regression test results
Top Management / Client	Deployment and Documentation Report	Hard copy	Angelo	May 31, 2026	Go-live confirmation, user manual, system documentation, handover checklist

4. Communication Methods and Guidelines

4.1 Formats Used

- Hard copy and Meeting: Used for high-stakes communications with management and the full team; ensures formal acknowledgment and discussion.
- E-mail: Used for technical updates between developers; provides a written trail and is efficient for asynchronous communication.
- Intranet / Shared Drive: Used for internal documentation accessible to all team members at any time.

4.2 Communication Frequency

- Weekly: Development progress reports sent every Monday during Phase 3 (March to May).
- Monthly: Formal status reports delivered on the 1st of each month to top management.
- Phase-based: Key deliverable documents distributed at each phase milestone (Feb 7, Mar 8, May 23, May 31).

4.3 Escalation Path

- Any unresolved issues or risks identified in weekly reports must be escalated to Angelo within 24 hours.
- Angelo is responsible for escalating critical risks to top management via the Monthly Status Report or an emergency hard-copy briefing.

5. Project Team Contact Directory

Name	Role	Phase Responsibility	Documents Owned	Format
Angelo	Team Leader / QA	Phase 1 and Phase 4 (Testing and Deployment)	Kick-off Doc, Scope Report, Test Reports, Deployment Report	Hard copy, Meeting
Abil	Developer: System Architecture	Phase 2 (Architecture) and Phase 3 (Finance)	System Architecture Doc, Financial Report (Sales), Cashflow Report	E-mail, Meeting
Meini	Developer: UI/UX and Analytics	Phase 2 (UI/UX) and Phase 3 (Analytics)	UI/UX Prototype, Dashboard Analytics, Export Reports	E-mail
Kennes	Developer: Database and Operations	Phase 2 (DB) and Phase 3 (Ops)	DB Design Doc, Functional Req., Inventory,	E-mail, Hard copy

			Supplier, Restock Modules	
Marvel	Developer: Core System	Phase 3 (Core modules)	Auth, POS, Receipt, Product Mgmt, P&L modules	E-mail

X. 5 Year Growth Plan SnapCash

Year 1 (2026): Product Validation & Market Entry

In the first year, SnapCash's primary focus is completing product development, conducting pilot testing, and achieving market validation. As a new startup, the company is not yet focused on profitability, but rather on ensuring that the product genuinely solves the challenges faced by MSMEs in managing transactions, inventory, and financial reporting.

The initial target is to acquire approximately 50–100 active MSME users in the Greater Jakarta area (Jabodetabek) through a free trial program, referrals, and direct outreach to small-scale F&B businesses.

Development Focus

- POS System refinement
- Inventory Management
- Financial Reporting
- Dashboard Analytics
- Basic Customer Support

Target KPIs

- 100 customers
- 70% retention rate
- 10 MSME partners as case studies
- Product-market fit achieved

Year 2 (2027): Regional Expansion

After achieving market validation, SnapCash begins expanding its reach to major cities such as Bandung, Surabaya, and Yogyakarta. The company's focus starts shifting from product development toward customer acquisition and improving user experience.

During this phase, the company begins building a sales and customer success team to assist MSMEs that are not yet familiar with digital systems during onboarding.

Development Focus

- Mobile Application
- QRIS Integration
- Multi-user Access
- Cloud Backup System

Target KPIs

- 300 active customers
- Churn rate < 8%
- 90% customer satisfaction
- Approaching break-even

Year 3 (2028): National Scaling

Entering the third year, SnapCash begins expanding to the national market. The product no longer functions solely as a cashier system, but evolves into a business decision-making tool through data analytics and basic automation.

The company also begins building partnerships with POS hardware distributors, MSME communities, and digital payment providers.

Development Focus

- AI-based Sales Forecasting
- Inventory Prediction
- Loyalty Program
- Multi-Outlet Management

Target KPIs

- 1,000 active customers
- 50 reseller partners
- Annual revenue > IDR 2 billion
- Stable profitability begins

Year 4 (2029): Operational Optimization

At this stage, the company has a sufficiently large customer base, so the focus shifts to improving operational efficiency and increasing customer lifetime value.

Rather than solely acquiring new customers, SnapCash begins offering premium features to increase revenue per customer.

Development Focus

- Accounting Integration
- Payroll Integration
- Supplier Management Enhancement
- Advanced Analytics

Target KPIs

- 2,000 active customers
- Retention > 85%
- Positive EBITDA
- Recognized as one of the leading local POS solutions among MSMEs

Year 5 (2030): Market Leadership in the MSME Segment

In the fifth year, SnapCash targets a position as one of Indonesia's leading POS and MSME business management providers. The company begins exploring partnership opportunities with banks, fintech companies, and MSME lending providers.

At this stage, growth comes not only from subscriptions, but also from additional services such as payment integration, premium analytics, and data-driven business consulting.

Development Focus

- AI Business Assistant
- Financial Insight Engine
- Marketplace Integration
- Open API Ecosystem

Target KPIs

- 3,000 active customers
- Annual revenue > IDR 8 billion
- Profit margin > 20%
- Preparation for ASEAN regional expansion

5-Year Financial Projection (Realistic)

Assumptions

Pricing Tiers:

- Basic: IDR 99,000/month
- Standard: IDR 199,000/month
- Premium: IDR 299,000/month

Average Revenue Per User (ARPU): IDR 150,000/month *(The majority of MSMEs tend to choose mid-tier or entry-level packages in the early stages.)*

Revenue Projection

Tahun	Pelanggan Aktif	Revenue
2026	100	Rp180.000.000
2027	300	Rp540.000.000
2028	1.000	Rp1.800.000.000
2029	2.000	Rp3.600.000.000
2030	3.000	Rp5.400.000.000

Operating Cost Projection

Tahun 1

Biaya masih tinggi karena development dan pemasaran awal.

Komponen	Biaya
Team Salaries	Rp180.000.000
Hosting & Infrastructure	Rp30.000.000
Marketing	Rp40.000.000
Operations	Rp20.000.000

Total	Rp270.000.000
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Tahun 2

Komponen	Biaya
Salaries	Rp240.000.000
Infrastructure	Rp50.000.000
Marketing	Rp80.000.000
Operasions	Rp30.000.000
Total	Rp400.000.000

Tahun 3

Komponen	Total
Human Resources	Rp480.000.000
Infrastructure	Rp120.000.000
Marketing	Rp180.000.000
Operasions	Rp70.000.000
Total	Rp850.000.000

Tahun 4

Komponen	Total
Human Resources	Rp900.000.000
Infrastructure	Rp180.000.000
Marketing	Rp300.000.000
Operasions	Rp120.000.000
Total	Rp1.500.000.000

Tahun 5

Komponen	Total
Human Resources	Rp1.200.000.000
Infrastructure	Rp250.000.000
Marketing	Rp500.000.000
Operasions	Rp250.000.000
Total	Rp2.200.000.000

Net Profit Projection

Tahun	Revenue	Cost	Net Profit
2026	Rp180.000.000	Rp270.000.000	(Rp90.000.000)
2027	Rp540.000.000	Rp400.000.000	Rp140.000.000
2028	Rp1.800.000.000	Rp850.000.000	Rp950.000.000
2029	Rp3.600.000.000	Rp1.500.000.000	Rp2.100.000.000
2030	Rp5.400.000.000	Rp2.200.000.000	Rp3.200.000.000

Profit Projection Explanation

This projection is more realistic as it assumes SnapCash will need the first year to build its customer base and conduct market validation. A first-year loss of IDR 90 million is still reasonable for a SaaS startup, as the company must invest in product development, marketing, and infrastructure before acquiring a sufficient number of customers.

Break-even occurs in the second year as the customer base grows and core development costs begin to decrease. From year three through five, profit growth is driven primarily by the nature of SaaS businesses, which carry relatively low incremental costs per additional customer. Therefore, even without aggressive customer growth, profit margins will continue to improve as the number of active customers increases. This is a more defensible scenario when presenting before lecturers, business case judges, or prospective investors, as the

growth assumptions remain conservative and aligned with real conditions in Indonesia's MSME market.